Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18,
Udyog Vihar, Gurugram, Haryana - 122018, ***Ph.***011-47623456

MM : 720 Botany 12th full syllabus VK **Time : 180 Min.**

Botany

1. Consider the following statements and choose the **correct** option
Statement A: Pollen grains lose viability within 30 minutes of their release in all the members of Leguminoseae, Rosaceae, and Solanaceae.
Statement B: Pollen grains can be stored in liquid nitrogen at –196°C for years.

(1) Only Statement A is correct
(2) Only statement B is correct
(3) Both statements A and B are correct
(4) Both statements A and B are incorrect

2. The ploidies of central cell and male gamete are respectively

(1) (n + n) and (n)
(2) (n) and (2n)
(3) (3n) and (n)
(4) (2n) and (2n)

3. Vegetative cell of male gametophyte in flowering plants

(1) Is smaller than the generative cell
(2) Has large irregularly shaped nucleus
(3) Floats in the cytoplasm of generative cell
(4) Has dense cytoplasm and is spindle shaped

4. The innermost layer of anther wall which surrounds the sporogenous tissue

(1) Has α-cellulosic fibrous bands on its cells
(2) Helps in dehiscence of anther
(3) Nourishes the developing pollen grains
(4) Has cells which lack nucleus

5. Outbreeding devices promote

(1) Self pollination
(2) Inbreeding depression
(3) Xenogamy
(4) Loss of genetic variation

6. Which of the following features is **not** related to the wind pollinated plants?

(1) Light and non-sticky pollen grains
(2) Presence of single ovule in each ovary
(3) Presence of nectaries
(4) Flowers having well exposed stamens

7. Read the following assertion and reason and choose the **correct** option.
Assertion : Cleistogamous flowers are invariably autogamous.
Reason : Geitonogamy is functionally cross-pollination involving a pollinating agent.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion
(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion
(3) Assertion is true statement but Reason is false
(4) Both Assertion and Reason are false statements

8. During double fertilisation.

(1) Two male gametes fuse with one egg cell
(2) Two polar nuclei fuse with two male gametes
(3) One male gamete fuses with both egg nucleus and polar nucleus
(4) One male gamete fuses with secondary nucleus and other with egg nucleus

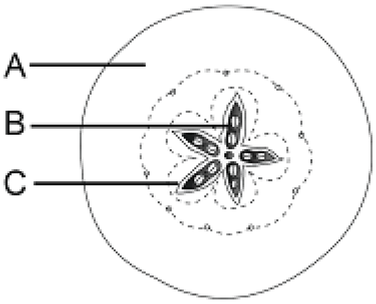
9. What would be the number of chromosomes in aleurone cell and egg cell respectively, of a monocot plant which has 20 chromosomes in its male gamete?

(1) 60 and 20
(2) 40 and 20
(3) 20 and 40
(4) 20 and 60

10. Parthenocarpy and apomixis are similar as both lack

(1) Fertilization process
(2) Embryo formation
(3) Fruit formation
(4) Development of seeds

11. Identify the given figure with **correct** labelling :



(1) True fruit of mango, A – Mesocarp

(2) True fruit of apple, B – Seed

(3) False fruit of strawberry, C – Epicarp

(4) False fruit of apple, B – Seed

12. Read the following statements and mark them as **True** (T) or **False** (F).
a. Coconut water from tender coconut has thousands of free nuclei.
b. Both endosperm and perisperm are similar in ploidy level and their origin.
c. Coleoptile is a hollow foliar structure that encloses leaf primordia in embryo.
d. Single, large shield shaped cotyledon of wheat embryo is known as epiblast.

	a	b	c	d
(1)	T	F	F	F
(2)	T	F	T	F
(3)	T	F	F	T
(4)	F	T	F	F

(1) (1)

(2) (2)

(3) (3)

(4) (4)

13. Female flowers of *Vallisneria* do **not** have

(1) Sticky stigma

(2) Unwettable stigma

(3) Nectaries

(4) Long stalk

14. Perisperm is

(1) Triploid tissue

(2) Remains of nucellus

(3) Another name of endosperm

(4) Found in pea seeds

15. Read the following statements and choose the **correct** option.
Assertion (A) : Pollen grains are well preserved as fossils.
Reason (R): Outer layer of pollen is made up of sporopollenin.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

16. Egg apparatus in angiosperms is

(1) 7 celled 8 nucleated

(2) 8 celled 7 nucleated

(3) 3 celled 3 nucleated

(4) 3 celled 8 nucleated

17. Which one is **correct** sequence of development of zygote into embryo?

(1) Pro-embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ globular embryo $\rightarrow$ Mature embryo

(2) Pro-embryo $\rightarrow$ Globular embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ Mature embryo

(3) Heart shaped embryo $\rightarrow$ Pro-embryo $\rightarrow$ Globular embryo $\rightarrow$ Mature embryo

(4) Globular embryo $\rightarrow$ Heart shaped embryo $\rightarrow$ Pro-embryo $\rightarrow$ Mature embryo

18. Apomictic embryos in many *Citrus* and mango varieties arise in the ovules, due to the protrusion and division of which of the following type of cells into the embryo sac?

(1) Egg Cell

(2) Synergids

(3) Nucellar cells

(4) Non-maternal tissue

19. Match the following columns and select the **correct** option.

	Column I		**Column II**
a.	Epiblast	(i)	Encloses embryonal root cap in monocots
b.	Coleorhiza	(ii)	A single cotyledon in monocots
c.	Scutellum	(iii)	Encloses leaf primordia in monocot embryo
d.	Coleoptile	(iv)	Remains of second cotyledon in some grasses

(1) a(iv), b(ii), c(iii), d(i)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(iv), b(i), c(ii), d(iii)

20. The phenomenon observed in few flowering plants such as some species of Asteraceae and Poaceae, in which seeds are produced without fertilisation is called

(1) Amphimixis

(2) Vegetative propagation

(3) Apomixis

(4) Sexual reproduction

21. Polyploid cells are seen in ___ layer of anther wall.

(1) Epidermis

(2) Tapetum

(3) Endothecium

(4) Middle layer

22. A type of Apomixis in which a diploid egg develops into a diploid embryo is seen in

(1) *Citrus*

(2) Apple

(3) Mango

(4) *Opuntia*

23. The central cell, after triple fusion, is called

(1) PEC (Primary endosperm cell)

(2) PEN (Primary endosperm nucleus)

(3) Zygote

(4) Endosperm

24. Filiform apparatus is a characteristic feature of

(1) Zygote

(2) Suspensor

(3) Egg

(4) Synergid

25. Embryo formation in *Citrus* **cannot** be a result of

(1) Double fertilization

(2) Apomixis

(3) Triple fusion

(4) Syngamy

26. Read the following assertion and reason and choose the **correct** option.
Assertion : Law of segregation is based on the fact that alleles do not show any blending.
Reason : Law of dominance is used to explain the expression of only one parental character in a monohybrid cross in the F_1.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

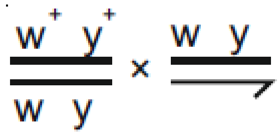
(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

27. Which of the given is not a test cross?

(1) Tt × tt

(2) RrYy × rryy

(3) 

(4) rryy × rryy

28. Match the column I with column II and select the **correct** option.

	Column I		Column II
a.	Pleiotropy	(i)	Human height
b.	Polygenic inheritance	(ii)	Phenylketonuria
c.	Incomplete dominance	(iii)	Blood group 'AB'
d.	Co-dominance	(iv)	Gene for starch grain size in pea

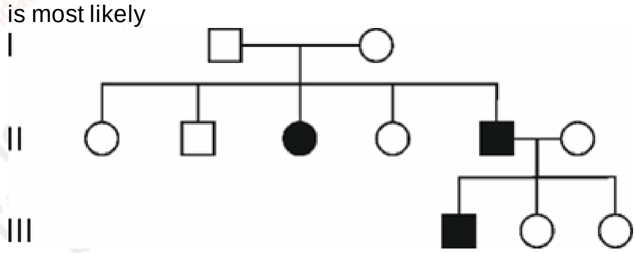
(1) a(ii), b(i), c(iv), d(iii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(iii), c(iv), d(i)

29. In the given pedigree chart, the trait under consideration is most likely



(1) Myotonic dystrophy

(2) Colour blindness

(3) Haemophilia

(4) Cystic fibrosis

30. A haemophilic male marries to a normal woman who is daughter of a haemophilic father. What percentage of male progenies of this couple will be normal?

(1) 50%

(2) 100%

(3) 25%

(4) 75%

31. Read the following statements and select the correct option.
Statement A: Dominance is not an autonomous feature of a gene.
Statement B: In co-dominance the F_1 generation resembles both parents.

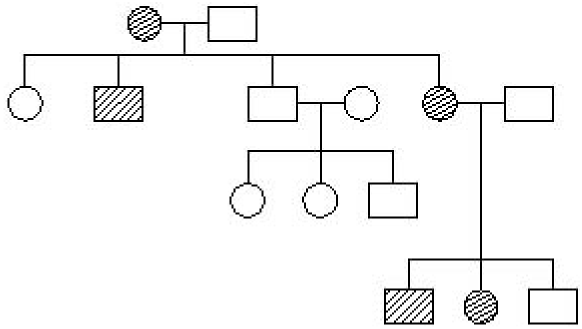
(1) Only statement A is correct

(2) Only statement B is correct

(3) Both the statements A and B are correct

(4) Both the statements A and B are incorrect

32. Given pedigree chart does **not** show the inheritance of trait like



(1) Autosomal recessive trait

(2) X-linked recessive trait

(3) Autosomal dominant trait

(4) X-linked dominant trait

33. Select the **incorrect** statement with respect to law of segregation

(1) The law is based on the fact that two factors of a character do not show any blending

(2) Both the traits are recovered as such in F_2 generation

(3) A gamete receives both the factors of a character during gamete formation

(4) Heterozygous individual produces two kinds of gametes each having one factor with equal proportion

34. Mark true **(T)** or false **(F)** for the statements given below and select the **correct** option.
(A) Starch synthesis in pea seeds is controlled by a pleiotropic gene
(B) Dominance is not an autonomous feature of a gene
(C) A single gene product never produce more than one effect

	A	**B**	**C**
(1)	T	T	F
(2)	F	T	T
(3)	T	F	F
(4)	F	F	T

(1) (1)

(2) (2)

(3) (3)

(4) (4)

35. In a family, father is having B blood group and mother is of A blood group. If their children show 50% probability of AB blood group, it indicates that

(1) Both father and mother are homozygous

(2) Both mother and father are heterozygous

(3) One of the parents is heterozygous

(4) Mother has a genotype $I^A I^A$ and father has genotype $I^A I^B$

36. The __A__ which represents the original phenotype is __B__ and the __C__ is generally the mutant type.
Fill in the blanks (A – C) by selecting the correct option.

(1) A- Unmodified allele, B- Dominant allele, C- Modified allele

(2) A- Wild allele, B- Dominant allele, C- Unmodified allele

(3) A- Modified allele, B- Functional allele, C- Mutant allele

(4) A- Functional allele, B- Wild allele, C- Unmodified allele

37. In DNA, substitution of purine base with a pyrimidine bas or vice versa is called

(1) Deletion

(2) Addition

(3) Transition

(4) Transversion

38. Phenylketonuria is an inborn error of metabolism in which

(1) Phenylalanine is converted to tyrosine

(2) Tyrosine is converted to phenylalanine

(3) Affected individual lacks phenylalanine hydroxylase enzyme

(4) Tyrosine get accumulated in brain resulting in mental retardation

39. Mark the **incorrect** match

(1) Down's Syndrome – Trisomy of chromosome 21

(2) Turner male – 44 + XO

(3) Thalassemia – Defect of α or β globin chain

(4) Klinefelter male – Have additional copy of X chromosome

40. Select the **incorrect** statement w.r.t. typical test cross.

(1) Progenies of such cross can be analysed to predict the genotype of the test organism

(2) Organism showing a dominant phenotype is crossed with the recessive parent

(3) Organisms are self-crossed

(4) Genotypic ratio is for monohybrid test cross will be 1 : 1

41. A : Deletion or insertion of a segment of DNA result in alteration in chromosomes.
R : Chromosomal aberrations are commonly observed in cancer cells.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

42. If mulattoes interbreed, what will be the ratio of the gradual gradation of skin colour phenotype in human beings (w.r.t. three pairs of polygens)?

(1) 1 : 4 : 6 : 4 : 1

(2) 1 : 6 : 15 : 20 : 15 : 6 : 1

(3) 1 : 2 : 2 : 4 : 1 : 2 : 1 : 2 : 1

(4) 9 : 3 : 3 : 1

43. Which of the following statements is **incorrect**?

(1) Individuals with different genotypes can have the same phenotype

(2) A genotype is the depiction of the alleles that represent an individual's genes

(3) Locus is the position of a gene on the chromosome

(4) A given gene can have only two alleles

44. Select the **incorrect** statement w.r.t. multiple allelism.

(1) Presence of more than two alleles for a gene

(2) Can be exemplified with ABO blood grouping in human beings

(3) Shows exception to Mendelian principles

(4) Can only be detected in an individual

45. In F_2 generation of Mendelian dihybrid cross (YYRR X yyrr)

(1) Yellow cotyledon and round seeded plants were obtained in 1 : 1 frequency

(2) Green cotyledon and round seeded plants were obtained in 3 : 1 frequency

(3) Round seeded plants obtained were 9

(4) All plants obtained were with yellow cotyledon and wrinkled seeded

46. In a clinical diagnosis a female 'A' was identified with absence of secondary sexual characters and has rudimentary ovaries. Which of the given statements can be **true** for the female 'A'?

(1) The chromosome complement of the female cannot be 44 + X0

(2) The female is suffering from a sex recessive linked disorder

(3) Mother of this female may be producing abnormal egg (22 + 0) and father producing normal sperm (22 + X)

(4) This female may pass this disorder to her daughters

47. Mark True **(T)** or False **(F)** for the given statements and select the **correct** option.
(A) Mendel selected 7 true-breeding pea plant varieties.
(B) His experiments had a small sampling size which gave greater credibility to the data that he collected
(C) Mendel for the first time applied statistical analysis and mathematical logic to problems in biology

	A	B	C
(1)	T	T	F
(2)	T	F	T
(3)	F	T	T
(4)	F	F	T

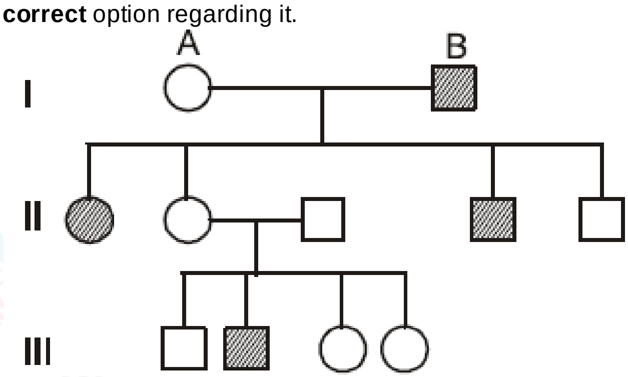
(1) (1)

(2) (2)

(3) (3)

(4) (4)

48. Study the following pedigree chart and choose the **correct** option regarding it.



(1) It is not applicable for an X-linked recessive disorder

(2) The genotype of A can be X^CX

(3) It shows inheritance of a Y-linked disorder

(4) The genotype of B cannot be aa

49. What is the probability of obtaining individuals with genotype PpqqRr, when a cross is made between the parents with genotypes PpQqRr × ppqqrr?

(1) $\frac{1}{8}$

(2) $\frac{1}{2}$

(3) $\frac{1}{16}$

(4) $\frac{1}{4}$

50. Post transcriptional modifications includes all, **except**

(1) Addition of 7mG cap

(2) Addition of poly A sequence

(3) Splicing of introns

(4) Addition of UTRs in m-RNA

51. An mRNA also has some additional sequences other than start, structural and stop codons which are not translated referred as untranslated regions (UTRs). Select **correct** statement w.r.t. UTRs
a. Needed for efficient translation
b. Present at both 5' - end after start codon and at 3' - end before stop codon
c. Present only at 5' - end of mRNA before capping
d. Present at 5' - end before start codon and at 3' - end after stop codon

(1) Only a and d are correct

(2) Only a and c are correct

(3) Only d is correct

(4) Only a is correct

52. Which of the following type of rRNA is **not** present in prokaryotic cell?

(1) 5 S

(2) 5.8 S

(3) 16 S

(4) 23 S

53. *lac* operon
(a) Has three structural genes
(b) Is repressible system
(c) Is anabolic system

(1) Only a is correct

(2) Only a and b are correct

(3) Only b and c are correct

(4) All a, b and c are correct

54. Which mRNA will translate to a polypeptide chain of 6 amino acids?

(1) AUGUUAAUAGACGAGUAACGAAUU

(2) AUGAUAGAGCAAUAGGACUUA

(3) AUGAUAUGACAAGACUAA

(4) AUGCCCAACCGUUAUUUAUAG

55. DNA was extracted from a bacterium. The proportion of cytosine was found to be 23%, then calculate the amount of adenine.

(1) 23%

(2) 27%

(3) 37%

(4) 17%

56. Prokaryotic transcription is similar to that of eukaryotic in

(1) Types of RNA polymerases

(2) Requirement of processing of m-RNA

(3) Being coupled with translation

(4) Direction of RNA formation

57. Match the columns and select the **correct** option.

	Column I		Column II
a.	Kornberg enzyme	(i)	Removal of introns
b.	DNA ligase	(ii)	Ribozyme
c.	Snurps	(iii)	Joining of Okazaki fragments
d.	Peptidyl transferase	(iv)	Removal of RNA primers

(1) a(iv), b(iii), c(ii), d(i)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(iii), c(i), d(ii)

(4) a(ii), b(i), c(iv), d(iii)

58. The DNA extraction was done from an *E. coli* culture, 60 min after its transfer from ^{15}N to ^{14}N medium. What will be the fraction of hybrid DNA in the total extracted DNA?

(1) $\frac{1}{4}$

(2) $\frac{1}{8}$

(3) $\frac{1}{2}$

(4) $\frac{1}{16}$

59. Find the **incorrect** statement.

(1) DNA polymorphism arises due to mutations

(2) The sensitivity of the DNA fingerprinting technique has increased by the use of PCR

(3) DNA polymorphisms are not inheritable from parents to offsprings

(4) Satellite DNA sequences that show very high degree of polymorphism is called VNTR

60. Which of the following statements are **true** regarding the experiment of Hershey and Chase with bacteriophage?
a. The unequivocal proof that DNA is the genetic material came from it.
b. Phage DNA was labelled with ^{32}P.
c. Proteins of phage contain phosphorus but no sulphur.
d. When bacteriophage was labelled with sulphur, no radioactivity was detected in supernatant

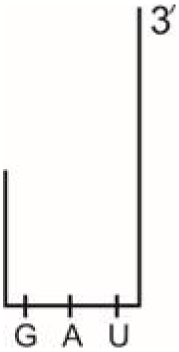
(1) a & b

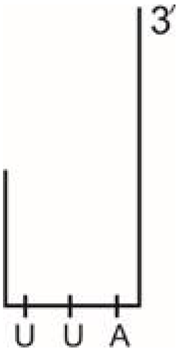
(2) b & c

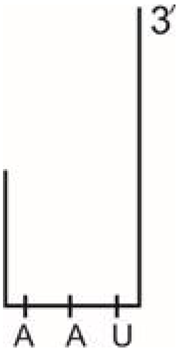
(3) c & d

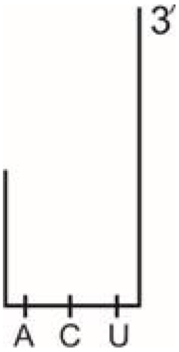
(4) d & a

61. mRNA with which of the following anticodon is **not** found?

(1) 

(2) 

(3) 

(4) 

62. Which of the given is/are involved in the process of DNA replication?
(A) Template DNA with polarity 3′ → 5′
(B) Template DNA with polarity 5′ → 3′
(C) DNA dependent DNA polymerase
(D) RNA dependent DNA polymerase

(1) Only (A) and (C)

(2) Only (C)

(3) (A), (B) and (C) only

(4) All (A), (B), (C) and (D)

63. Select the **incorrect** statement w.r.t. tRNA.

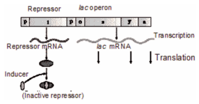
(1) It has an anticodon loop that has bases complementary to the codon of mRNA

(2) It has an amino acid acceptor end to which it binds to amino acid

(3) tRNAs are specific for each amino acid

(4) The secondary structure of tRNA looks like inverted L

64. Examine the figure given below and select the **correct** match



(1) 'i' – Constitutive expression

(2) 'z' - Produce β-galactoside

(3) 'a' - For permease synthesis

(4) 'o' – Initial binding site for RNA polymerase

65. In a transcription unit, the promoter is located towards.

(1) 5' end of the template strand

(2) 5' end of the coding strand

(3) 3' end of the non-template strand

(4) 3' end of the sense strand

66. In Griffith’s experiments, mice died when injected with

(1) Heat killed R-strain bacteria

(2) Live R-strain bacteria

(3) Heat killed S-strain combined with live R-strain bacteria

(4) Heat killed S-strain bacteria

67. Discontinuous fragments of DNA which are formed during replication of DNA are joined by

(1) DNA ligase

(2) Lipase

(3) DNA polymerase

(4) RNA polymerase

68. Genetic material in λ-phage have

(1) 5386 bp long DNA

(2) 48502 nucleotides

(3) 48502 nucleotide pairs

(4) 5386 nucleotides

69. RNA polymerase I transcribes all types of rRNAs **except**

(1) 28S rRNA

(2) 18S rRNA

(3) 5.8S rRNA

(4) 5S rRNA

70. Polynucleotide phosphorylase (Severo ochoa) enzyme

(1) Is capable of polymerising DNA

(2) Work in template independent manner

(3) Work in template dependent manner

(4) Both (A) and (C)

71. Polynucleotide phosphorylase (Severo ochoa) enzyme

(1) Is capable of polymerising DNA

(2) Work in template independent manner

(3) Work in template dependent manner

(4) Both (1) & (3)

72. O. Avery, C. MacLeod and M. McCarty worked to determine the biochemical nature of transforming principle, related to Griffith's experiment. Their work led to conclude that

(1) Transforming substance is RNA

(2) DNA is the hereditary material

(3) Transforming substance is protein

(4) Enzyme proteases and RNases affect transformation

73. How many amino acid(s) will be there in the polypeptide coded by the mRNA with sequence 5' AUGUACAAGUAG 3' ?

(1) Three

(2) Two

(3) One

(4) Four

74. In a dsDNA sample, out of the total nitrogenous bases, guanine was found to be 18%. What would be the percentage of other bases in this sample?

	C	T	A
(1)	18%	32%	18%
(2)	18%	18%	32%
(3)	18%	32%	32%
(4)	32%	18%	18%

(1) (1)

(2) (2)

(3) (3)

(4) (4)

75. Arrange the following steps involved in DNA fingerprinting in the **correct** order and choose the option accordingly.
P – Electrophoresis
Q – Autoradiography
R – Hybridisation with VNTR probe
S – Blotting
T – Isolation of DNA

(1) P → Q → S → T → R

(2) T → P → S → R → Q

(3) T → S → P → Q → R

(4) Q → R → S → T → P

76. A : Ladybird is useful in controlling aphids.
R : Dragonflies are useful to get rid of mosquitoes.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

77. Vodka has approximately __(i)__ concentration of alcohol whereas wine has __(ii)__.

	(i)	(ii)
(1)	30 – 40%	40 – 50%
(2)	9 – 12%	60 – 70%
(3)	60 – 80%	9 – 12%
(4)	40 – 45%	70 – 80%

(1) (1)

(2) (2)

(3) (3)

(4) (4)

78. 'Toddy', a traditional drink of some parts of Southern India is made by fermenting sap from spadices of

(1) *Caryota urens*

(2) *Glycine max*

(3) Bamboo shoots

(4) Wheat

79. In fermentation of dough, cheese making and production of beverages, the main gas produced is

(1) Methane and H_2

(2) Carbon dioxide

(3) Carbon monoxide

(4) Methane, CO_2 and H_2

80. Find the **correct** sequence of stages during treatment of sewage water upto the stage when effluent is disposed in river.
(i) Filtration of floating debris
(ii) Tertiary treatment
(iii) Lowering of BOD
(iv) Anaerobic digestion of sludge
(v) Introduction of inoculum into aeration tank

(1) (i) → (v) → (iii) → (iv)

(2) (i) → (v) → (iii) → (ii) → (iv)

(3) (iii) → (i) → (v) → (ii) → (iv)

(4) (i) → (iii) → (v) → (iv) → (ii)

81. A bacterium named as *Propionibacterium sharmanii* produces CO_2, which is responsible for the large holes in

(1) Roquefort cheese

(2) Camembert cheese

(3) Swiss cheese

(4) Both (A) & (C)

82. Cyclosporin A, used as an immunosuppressive agent in organ-transplant patients, is produced by the fungus

(1) *Monascus purpureus*

(2) *Trichoderma polysporum*

(3) *Aspergillus niger*

(4) *Mortierella renispora*

83. Match the items in column-I with column-II and select the most suitable option.

Column-I	Column-II
a. Baculoviruses	(i) Check disease causing microbes
b. Flocs	(ii) Found in anaerobic sludge
c. Methanogens	(iii) Aerobic microbes
d. Lactic acid bacteria	(iv) Attack insects and other arthropods

(1) a(iv); b(iii); c(ii); d(i)

(2) a(iv); b(ii); c(iii); d(i)

(3) a(i); b(iv); c(ii); d(iii)

(4) a(ii); b(i); c(iii); d(iv)

84. A : *Saccharomyces cerevisiae* is used for commercial production of ethanol.
R : *Aspergillus niger* is the producer of citric acid.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

85. A : Antibiotics are chemical substances produced by some microbes, which can kill or retard growth of other microbes.
R : Penicillin was the first antibiotic to be discovered.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

86. Which of the following statements is **incorrect**?

(1) Biogas can be used as source of energy as it is inflammable

(2) Bacteria which grow anaerobically on cellulosic material produce methane, CO_2 and H_2

(3) Methanogens are found in aerobic sludge during STP

(4) Dung of cattle is used for generating gobar gas

87. Statin acts by competitively inhibiting the enzyme responsible for synthesis of

(1) Citric acid

(2) Lactic acid

(3) Acetic acid

(4) Blood cholesterol

88. Curd is more nutritious than milk as it is rich in number of vitamins specially

(1) B_{12}

(2) A

(3) K

(4) D

89. Which of the following can be biofertilisers?
(a) Rhizobium
(b) Glomus
(c) Azotobacter
(d) Cyanobacteria

(1) Only (a) and (b)

(2) Only (a) and (c)

(3) Only (b) and (d)

(4) All (a), (b), (c) and (d)

90. Match the Column-I and Column-II w.r.t. bioactive molecules and their importance.

Column-I	Column-II
(i) Lipase	(P) Blood-cholesterol lowering agents
(ii) Streptokinase	(Q) Detergent formation
(iii) Statins	(R) Degrade starch
(iv) Amylase	(S) Clot buster

	(i)	(ii)	(iii)	(iv)
(1)	S	R	Q	P
(2)	Q	S	P	R
(3)	Q	P	S	R
(4)	R	P	Q	S

(1) (1)

(2) (2)

(3) (3)

(4) (4)

91. During secondary sewage treatment, mesh like structure is formed, known as flocs. It is

(1) Masses of bacteria associated with fungal filaments

(2) Combination of algae and fungi

(3) Mass of bacteria only

(4) Mass of fungi only

92. Aeration tanks show the vigorous growth of

(1) Harmful anaerobic microbes

(2) Useful aerobic heterotrophic microbes

(3) Useful aerobic chemotrophic microbes

(4) Phototrophic, oxygenic bacteria

93. Find the **mismatched** pair between microbes and its products

	Microbes	Product
(1)	*Aspergillus niger*	Citric acid
(2)	*Acetobacter aceti*	Acetic acid
(3)	*Streptococcus*	Streptokinase
(4)	*Lactobacillus*	Amylase

(1) (1)

(2) (2)

(3) (3)

(4) (4)

94. All solids that settle during primary stage of waste water treatment forms

(1) Primary sludge

(2) Primary effluent

(3) Activated sludge

(4) Flocs

95. Read the given statements and select the **correct** option.
Statement A: Ministry of Environment and Forest has initiated Ganga Action Plan.
Statement B: Effluent from secondary treatment cannot be released into water bodies.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statement A & B are correct

(4) Both statement A & B are incorrect

96. The grit present in sewage is removed by

(1) Sequential filtration

(2) Sedimentation

(3) Biological treatment

(4) Tertiary treatment

97. Biogas produced in anaerobic sludge digesters is a mixture of

(1) Methane and carbon dioxide only

(2) Hydrogen sulphide and methane only

(3) Methane, carbon dioxide and hydrogen sulphide

(4) Carbon dioxide, methane and oxygen

98. Which of the following is **not** a tool or component, successfully used for integrated pest management in several crops?

(1) Higher use of pesticides

(2) Summer ploughing

(3) Use of parasites

(4) Use of pest resistant plant varieties

99. Choose the **incorrect** match.

(1) Triangular age pyramid – Expanding population

(2) Tiger census – Often based on pug marks and fecal pellets

(3) Bell shaped age pyramid – Mature population

(4) Urn shaped age pyramid – Positive growth

100. Growing population

(1) Has no post-reproductive individuals

(2) Has more pre-reproductive individuals than reproductive ones

(3) Show urn shaped age pyramid

(4) Is also said to be mature population

101. Read the following statements and choose the option which is true for them.
Statement-A : All the mammals are able to maintain homeostasis, so they never hibernate.
Statement-B : Majority of animals are unable to maintain a constant internal environment.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

102. Choose the **correct** option regarding given statements.
Statement A: Polar aquatic mammals like seals have a thin layer of fat (blubber) below their skin.
Statement B: Polar aquatic mammals are not able to maintain homeostasis.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are correct

(4) Both statements are incorrect

103. "Mammals from colder climates generally have shorter ears and limbs to minimise heat loss". This is called

(1) Allen's rule

(2) Jordan's rule

(3) Rensch's rule

(4) Bergmann's rule

104. A : Herbivores and plants are not affected adversely by competition.
R : Predators in nature are never prudent and they over exploit the prey.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

105. A : Altitude sickness can be experienced at high altitude where body doesn't get enough oxygen due to high atmospheric pressure.
R : Under these conditions body increases RBC production, decrease binding capacity of haemoglobin and decrease breathing rate.

(1) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(2) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(3) Both Assertion and Reason are false statements

(4) Assertion is true statement but Reason is false

106. A : Antarctic fish can survive below 0°C
R : The body fluid of these fishes contains antifreeze solutes.

(1) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(2) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(3) Both Assertion and Reason are false statements

(4) Assertion is true statement but Reason is false

107. In how many of the given interactions, at least one species is harmed?

Mutualism, Competition, Predation, Parasitism, Commensalism, Amensalism, Protocooperation

(1) Three

(2) Four

(3) Five

(4) Two

108. Ecological niche of an organism includes all, **except**

(1) Range of conditions it can tolerate

(2) Resources it utilises

(3) Body organisation which it has

(4) Function it performes in an ecological system

109. Emigration is

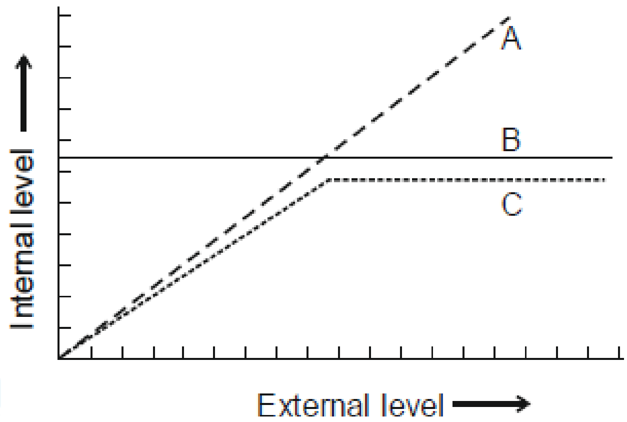
(1) Number of births during a given period in the population that are added to the initial density

(2) Number of deaths in the population during a given period

(3) Number of individuals of the same species that have come into the habitat from elsewhere during the time period under consideration

(4) Number of individuals of the population who had left the habitat and gone elsewhere during the time period under consideration

110. Consider the following diagrammatic representation of organismic response.



Which of the following organisms show the response to abiotic factors that is represented by the line 'A'?

(1) Plants

(2) Humans

(3) Birds

(4) Mammals

111. Exponential growth does not show

(1) Asymptote condition

(2) J-shaped growth curve

(3) Geometric fashion of growth

(4) $\frac{dN}{dt} = rN$ relation

112. A sedentary sea anemone gets attached to the shell lining of hermit crab. The association is

(1) Amensalism

(2) Commensalism

(3) Protocooperation

(4) Ectoparasitism

113. Which among the following is **true** for the Urn-shaped age pyramids?

(1) Have large number of pre-reproductive individuals than reproductive ones

(2) It shows growing population

(3) It shows negative growth of the population

(4) It show stable population

114. Siberian Crane usually visit India in winter at

(1) Bharatpur sanctuary

(2) Kanha National park

(3) Periyar sanctuary

(4) Chilka lake

115. The equation $\frac{dN}{dt} = rN\left(\frac{K-N}{K}\right)$ describes the Verhulst-Pearl logistic growth of population. In this equation K represents

(1) Available nutrient resources

(2) Population density at a particular time

(3) Intrinsic rate of increase in population

(4) Carrying capacity

116. (i) Small animals have a larger surface area relative to their volume.
(ii) About 99 percent of animals cannot maintain a constant internal environment.
(iii) Diapause, a stage of suspended development is seen in snails and fishes.
(iv) Thermoregulation is energetically expensive for many organisms.
(v) Conformers had not evolved to become regulators.
Choose the **correct** option.

(1) All the statements are correct

(2) Except (iii) all are correct

(3) (v), (iii), (iv) are correct

(4) (i), (iii), (v) are correct

117. Gause's principle is related to

(1) Competitive exclusion

(2) Population density

(3) Competitive co-existance

(4) Resource partitioning

118. The organisms in which body temperature changes with the ambient temperature are

(1) Birds

(2) Humans

(3) Mammals of colder region

(4) Plants

119. Which among the following attributes are observed in the desert plants, *Opuntia*?
a. Thick cuticle on its leaf surface
b. Presence of spines
c. Shows crassulacean acid metabolism
d. Photosynthesis performed by flattened stem

(1) All except a

(2) b and c only

(3) a, b, c and d

(4) All except c

120. A population is growing in a habitat which has limited resources. This population will show the sequence of growth phases

(1) Lag phase → Log phase → Crash

(2) Log phase → Lag Phase → Asymptote

(3) Lag phase → Deceleration → Log phase → Acceleration

(4) Lag phase → Acceleration → Deceleration-→ Asymptote

121. Which of the following pairs of organism breed only once in their life time?

(1) Most Birds, Mammals

(2) Most Birds, Pacific salmon fish

(3) Bamboo, Pacific salmon fish

(4) Mammals, Bamboo

122. The most important ecologically relevant abiotic environmental factor is

(1) Air

(2) Light

(3) Temperature

(4) Soil

123. Ecologically, standing crop means the amount of living material present in

(1) Trophic level of green plants only at a given time

(2) Same trophic levels of only consumers per unit area

(3) Different trophic levels at a given time per unit area

(4) Same trophic level of only primary consumers per unit area

124. Find the **odd** one out w.r.t. ecosystem function.

(1) Productivity

(2) Decomposition

(3) Stratification

(4) Energy flow

125. Anthropogenic ecosystem possesses all of the given characteristics, **except**

(1) Have little diversity

(2) Self regulatory mechanism

(3) Simple food chain

(4) Little cycling of nutrients

126. In most ecosystems, all the pyramids of number, energy and biomass are upright. It indicates that
(a) Producers are more in number and biomass than the herbivores.
(b) Herbivores are less in number and biomass than the carnivores.
(c) Energy at a lower trophic level is always more than at a higher level.
Choose the **correct** option.

(1) Only (a) is correct

(2) Only (b) is correct

(3) Both (a) and (c) are correct

(4) Both (b) and (c) are correct

127. Which of the given ecological pyramids is **not** upright?

(1) Pyramid of number in tree ecosystem

(2) Pyramid of number in grassland

(3) Pyramid of energy in pond

(4) Pyramid of biomass in grassland

128. How many of the following can be considered as least productive ecosystem?
Tropical rain forest, Desert, Sugarcane field, Coral reef, Grassland, Deep sea

(1) Four

(2) Three

(3) One

(4) Two

129. In a stable ecosystem such as tropical rain forest, which of the following is a limiting factor?

(1) Sunlight

(2) Water

(3) Availability of nutrients

(4) O_2 concentration

130. Read the following statements and select the **correct** one(s).
a. A trophic level represents a functional level not a species as such.
b. A given species may occupy more than one trophic levels in the same ecosystem at the same time.
c. Energy at a lower trophic level is always less than at a higher level

(1) Only a

(2) Only a and b

(3) Only b and c

(4) All a, b and c

131. A : All animals depend on plants directly or indirectly for their food needs.
R : The consumers that feed on herbivores are called carnivores.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

132. Stratification is

(1) Vertical distribution of different species occupying different levels

(2) Horizontal distribution of different species along the equater of earth

(3) Invasion of a new species in a particular area

(4) Identification and classification of biological species.

133. The annual net primary productivity in dry weight of organic matter of the whole biosphere is approximately

(1) 170 billion tons

(2) 55 billion tons

(3) 115 billion tons

(4) 80 billion tons

134. A : Ecology is the study of relationships of living organisms with the physio-chemical factors and other species of their environment.
R : In nature populations of different species in a habitat do not interact and live in isolation.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion

(3) Assertion is true statement but Reason is false

(4) Both Assertion and Reason are false statements

135. In forest ecosystem, bottom strata or layer is occupied by

(1) Trees only

(2) Herbs only

(3) Both shrubs and trees

(4) Both herbs and grasses

136. Select the **odd** one w.r.t anthropogenic ecosystems.

(1) Lack of self regulatory mechanisms

(2) Little cycling of nutrients

(3) Very low productivity

(4) Presence of simple food chain

137. If 15 J of energy is trapped at producer level, then how much energy will be available to tertiary consumer (T_4 level)?

(1) 0.015 J

(2) 0.0015 J

(3) 0.15 J

(4) 0.00015 J

138. The relation between net primary productivity **(NPP) and gross primary productivity (GPP) is** represent**ed by**

(1) NPP = GPP – R

(2) NPP = GPP + R

(3) NPP = GPP × R

(4) NPP = $\frac{GPP}{R}$

139. Which of the following is **not** the feature of humus?

(1) Dark coloured

(2) Amorphous and colloidal

(3) Serves as reservoir of nutrients

(4) Least resistant against microbial action

140. Study the given food chain
Dead organic matter → Earthworm → A → Falcon
Choose the **correct** option for organism A.

(1) Lice

(2) Sparrow

(3) Wolf

(4) Zooplanktons

141. Spindle shaped pyramid of number is found in ecosystem of

(1) Grassland

(2) Tree

(3) Pond

(4) River

142. Zooplanktons in aquatic ecosystem are

(1) Producers

(2) Saprotrophs

(3) Primary consumers

(4) Tertiary consumers

143. The rate of decomposition of detritus is slow when

(1) Temperature is more than 45°C

(2) It contains carbohydrates

(3) There is optimum moisture

(4) There is good aeration

144. In ecological pyramids, the saprophytes

(1) Appear at second trophic level

(2) Appear at third trophic level

(3) Are not given any place

(4) Are included at first trophic level with plants

145. Primary productivity of an area depends on
a. Variety of environmental factors
b. Availability of nutrients
c. Plant species inhabiting that area
d. Photosynthetic capacity of plants
Choose the **correct** option.

(1) Only a and b

(2) Only c and d

(3) Only b, c and d

(4) All a, b, c and d

146. Consider the given assertion and reason, and choose the **correct** option.
Assertion: The pyramid of biomass in sea is generally inverted.
Reason: The biomass of fishes far exceeds that of phytoplankton.

(1) Both the assertion and reason are true and the reason is the correct explanation of assertion.

(2) Both the assertion and reason are true but the reason is not the correct explanation of assertion.

(3) Assertion is true but the reason is false statement.

(4) Both assertion and reason are false statements.

147. Which of the following is **incorrect** w.r.t. ecosystem?

(1) Ecosystems are not exempt from the second law of thermodynamics

(2) Herbaceous and woody plants are major producers in terrestrial ecosystem

(3) Sun is the only source of energy for all ecosystems on Earth

(4) The green plants in the ecosystem are called producers

148. Edaphic factor of an ecosystem includes

(1) Humidity of air

(2) Physical and chemical properties of soil

(3) Slope of earth surface

(4) Wind and air current at that place

149. According to the 'Rivet popper hypothesis' airplane represents the

(1) Extinct species

(2) Ecosystem

(3) Alien species

(4) Biodiversity

150. Consider the following equation
log S = log C + Z log A
In the given species-area relationship equation, C represents

(1) Species richness

(2) Y-intercept

(3) Area

(4) Regression coefficient

151. "When we conserve and protect the whole ecosystem, its biodiversity at all levels is protected". This approach includes all of the following, **except**

(1) Biosphere reserves

(2) Seed banks and wildlife safari parks

(3) National parks

(4) Sanctuaries

152. Biodiversity decreases as we move from

(1) Temperate to tropical areas

(2) Poles to equator

(3) High to low altitude

(4) Colombia to Greenland

153. A population of *Rauwolfia vomitoria* is adapted to different Himalayan ranges and shows variation in potency and concentration of its active chemical. This way it exhibits

(1) Species diversity

(2) Ecological diversity

(3) Genetic diversity

(4) Community diversity

154. Which of the given is an example of organisms with recent extinction in Mauritius?

(1) Dodo

(2) Quagga

(3) Thylacine

(4) Steller's sea cow

155. The most important cause driving animals and plants to extinction is

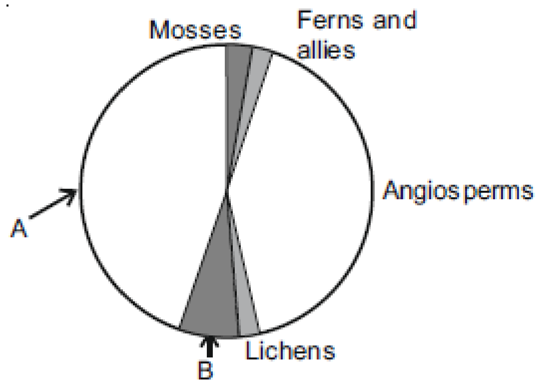
(1) Habitat loss and fragmentation

(2) Alien species invasions

(3) Over-exploitation

(4) Co-extinctions

156. Following pie-chart represents proportionate number of species of major taxa of plants. Select the option which **correctly** represents the labels A and B.



	A	B
(1)	Gymnosperms	Fungi
(2)	Algae	Gymnosperms
(3)	Algae	Fungi
(4)	Fungi	Algae

(1) (1)

(2) (2)

(3) (3)

(4) (4)

157. Loss of biodiversity in a region may lead to

(1) Greater resistance to drought

(2) Decline in plant production

(3) Decreased variability in certain ecosystem processes

(4) Decreased pest and disease cycles

158. Loss of biodiversity in a region may lead to all, **except**

(1) Decline in plant productivity

(2) Reduced resistance to environmental perturbations

(3) Increased variability of ecosystem processes

(4) Decreased variability in pest and disease cycle

159. The records or estimates regarding biodiversity do **not** give any figures for prokaryotes because
(a) Conventional taxonomic methods are not suitable for identifying microbial species.
(b) Many prokaryotic species are simply not culturable under laboratory conditions.
(c) They are not seen with naked eyes.
(d) They do not show sexual reproduction which is essential to increase the biodiversity.
The **correct** ones are

(1) a & b

(2) b & c

(3) c & d

(4) a, b & d

160. Which of the following is an example of organism with recent extinction in Africa?

(1) Dodo

(2) Quagga

(3) Thylacine

(4) Steller's sea cow

161. The historic convention the Earth Summit held in 1992 was called to take appropriate measures for

(1) Conservation of biodiversity and sustainable utilisation of its benefits

(2) Reducing greenhouse gas emission and global warming

(3) Reducing release of persistent organic pollutant in atmosphere

(4) *Ex situ* conservation of endangered species of wild flora and fauna

162. **Assertion (A) :** Species with small body size are more susceptible to extinction.
Reason (R) : Species which are more susceptible to extinction are mostly at lower trophic levels in food chain.
In the light of the above statements, select the **correct** option.

(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

163. Number of animal species present in India are nearly

(1) 45,000

(2) 90,000

(3) 3,00,000

(4) 1,00,000

164. Select the **incorrect** match.

(1)	*Ex situ* conservation	–	Zoological park
(2)	Co-extinctions	–	Plant-pollinator mutualism
(3)	Over-exploitation	–	Extinction of passenger pigeon
(4)	Sixth extinction	–	Purely non-anthropogenic

(1) (1)

(2) (2)

(3) (3)

(4) (4)

165. Select the **incorrect** match.

(1) Khasi and Jaintia hills – Meghalaya

(2) Bastar – Rajasthan

(3) Sarguja – Madhya Pradesh

(4) Western ghats – Karnataka and Maharashtra

166. How many of the following is/are the key criteria used for determining a hot spot?
(a) Low biodiversity or species richness
(b) Degree of threat measured in terms of habitat loss
(c) High degree of endemism
(d) Production of economically beneficial materials like food, fibre, drugs etc.

(1) One

(2) Two

(3) Three

(4) Four

167. We should conserve our biodiversity for which of the following reasons?
a) Economic benefits
b) Ecosystem services
c) Some ethical reasons
The **correct** one(s) is/are

(1) Only a

(2) Only a & b

(3) Only b

(4) All a, b & c

168. Among animals $\underline{\quad a \quad}$ are most species rich taxonomic group making more than $\underline{\quad b \quad}$ of the total.

(1) a-Molluscs
b-70%

(2) a-Insects
b-70%

(3) a-Crustaceans
b-50%

(4) a-Insects
b-40%

169. The extinction of Steller's sea cow occurred as a result of

(1) Alien species invasion

(2) Co-extinction

(3) Over-exploitation by humans

(4) Habitat loss and fragmentation

170. Exploring molecular, genetic and species level biodiversities for products of economic importance is called

(1) Bioprospecting

(2) Bioremediation

(3) Biocontrol

(4) Biodiversity

171. Choose the **incorrect** match regarding example of recent extinctions

(1) Thylacine – Australia

(2) Dodo - Myanmar

(3) Steller's Sea cow- Russia

(4) Quagga – Africa

172. On a logarithmic scale the correct representation of species-area relationship is represented by the equation

(1) Log S = Log C – Z Log A

(2) Log S = Log C + Z Log A

(3) Log A = Log C + Z Log S

(4) Log S = Log A + Z Log C

173. Presently how much percentage of mammal species are facing the threat of extinction?

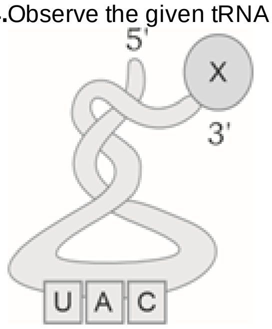
(1) 12%

(2) 23%

(3) 31%

(4) 32%

174. Observe the given tRNA



Select the option that cannot be true regarding the above tRNA?

(1) This tRNA will bind at AUG codon of mRNA

(2) This tRNA binds with amino acid methionine at its 3' end

(3) The loop with UAC is anticodon loop of this tRNA

(4) The amino acid X would be the same when anticodon UAC is changed to UAA

175. Genetic code is degenerate, which means

(1) One codon codes for only one amino acid

(2) Codon is read on mRNA in a contiguous fashion

(3) Three codons do not code for any amino acid

(4) Some amino acids are coded by more than one codon

176. 3D structure of tRNA is

(1) Clover-leaf like

(2) Inverted L-shaped

(3) J shaped

(4) V shaped

177. Split gene arrangement is a feature of

(1) *E. coli*

(2) *Streptococcus*

(3) Plant cell

(4) *Mycoplasma*

178. Which of the following statements is **false** w.r.t. phenylketonuria?

(1) It is an inborn error of metabolism

(2) Along with accumulation of the derivatives of phenylalanine these are also excreted through urine

(3) Affected individual lacks phenylalanine hydroxylase enzyme

(4) Phenylalanine is converted into tyrosine that when accumulated in brain results in mental retardation

179. Four genes a, b, c and d are present on a chromosome. Percentage of recombination between a and b = 20%; a and d = 2%, c and b = 8% and d and b = 18%. What will be the sequence of genes on the chromosome?

(1) adcb

(2) cdab

(3) dcab

(4) acbd

180. In which of the given population interactions one species harm other without getting benefited?

(1) *Penicillium* and bacteria

(2) Tiger and deer

(3) *Paramecium aurelia* and *P.caudatum*

(4) Barnacles and whale